

Dashlin Naidoo

Software Developer | Computer Science and GeoInformatics

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PROFILE

Early-career software developer and BSc GeoInformatics: Computer Science graduate, currently completing an Honours degree at Stellenbosch University. Experienced in Python, Java, C, React, .NET, .NET MAUI, Firebase and Assembly, with project work across full-stack development, algorithms, game AI, compiler construction, machine learning and geospatial data science. Builds practical, maintainable applications, learns quickly across new tools and communicates technical ideas clearly through tutoring and project work.

TECHNICAL SKILLS

Languages: Python, Java, C, Assembly, JavaScript, HTML, CSS

Frameworks and Platforms: React, .NET, .NET MAUI, Firebase Authentication, npm

Software Concepts: Object-oriented programming, data structures, algorithms, compiler basics, client-side routing, state management, authentication, local caching, low-connectivity optimisation

Data and GIS: Machine learning, regression, classification, clustering, geospatial analysis, satellite-derived vegetation indices

Design and Tools: Figma, UI/UX design, academic research, report writing, presentation

WORK EXPERIENCE

Computer Science Tutor

Magane CS Tutors | Present

- Helped students build Python and Java programming skills by explaining core syntax, debugging errors and guiding practical coding exercises.
- Improved learner confidence in computer science concepts by breaking down problem-solving, object-oriented programming and data structures into step-by-step examples.
- Strengthened communication and mentoring skills by adapting explanations to different learning levels in a dedicated computer science tutoring environment.

Lecture Assistant

Stellenbosch University | Jan 2024 - Present

- Improved student understanding across GIT 124, GIT 214, GIT 241 and GIT 342 by explaining technical concepts, guiding practical work and supporting first-, second- and third-year students.
- Strengthened practical learning outcomes for students at different academic levels by providing troubleshooting support, concept explanations and step-by-step coursework guidance.
- Balanced tutoring and academic responsibilities by preparing practical support, responding to student needs and communicating technical ideas clearly.

Internship

POS IT Excellence | Jul 2023 - Present

- Contributed to professional back-end development and design tasks by applying academic programming knowledge to real development requirements.
- Improved practical software development capability by completing back-end coding tasks, collaborating in a professional environment and working to delivery expectations.
- Built stronger accountability and teamwork habits by contributing to development work under workplace standards and adapting to real project requirements.

Student Developer / Academic Projects

Stellenbosch University | Jan 2023 - Dec 2025

- Built multiple software applications and technical projects using Python, Java, C, .NET, .NET MAUI and Assembly by applying computer science principles across coursework and project work.
- Developed user-focused application designs by using Figma for UI/UX prototyping and translating interface concepts into structured software projects.
- Strengthened full-stack, systems programming and data-focused development skills by completing projects involving React, Firebase Authentication, local caching, game AI, compiler construction, sockets and geospatial analysis.
- Improved software problem-solving ability by implementing algorithms, debugging application logic and applying data structures across academic and applied development projects.

SELECTED PROJECTS

Routing Application for Gendac

Technologies: .NET, .NET MAUI, Firebase Authentication

- Developed an Android-focused routing application from the ground up by implementing secure Firebase Authentication, email/password and social login options, token-based sessions and local caching for low-connectivity field environments.

Personal Website

Technologies: React, npm, client-side routing

- Built a responsive personal website by using component-driven React development, reusable components, props, hooks, package management and single-page navigation.

Othello Simulator

Technologies: C, alpha-beta pruning, game AI

- Engineered an intelligent Othello simulator by implementing recursive decision-tree traversal, heuristic board-state evaluation and alpha-beta pruning to reduce computational overhead.

Compiler

Technologies: C, lexical analysis, parsing

- Built a compiler from scratch in C by applying lexical analysis, parsing logic, low-level systems programming and memory-aware implementation practices.

Braille to Text Translator

Technologies: Java, deterministic finite automata, pattern recognition

- Built a converter using deterministic finite automata to map complex character sets and handle special Braille formatting rules, strengthening state management and pattern recognition.

Multiplayer Chess Project

Technologies: Python, sockets, game logic

- Built a real-time multiplayer chess game by implementing move validation, check/checkmate detection and network-based turn synchronisation.

EDUCATION

BScHons: GeoInformatics

Stellenbosch University | Jan 2026 - Dec 2026

- Thesis: Predicting Clinic/Ward-Level Tuberculosis Notification Risk Using Machine Learning and Geospatial Analysis. Coursework includes Python-based regression, classification, clustering and geospatial data science.

BSc: GeoInformatics: Computer Science

Stellenbosch University | Jan 2023 - Dec 2025

- Completed coursework in Geographic Information Technology and Computer Science, including Java, C, Assembly, data structures, algorithms, UI/UX design and geospatial data science.
- Presented an article at the International Conference of Military GeoScience 2024.

ADDITIONAL STRENGTHS

- Clean coding, debugging and problem-solving
- Object-oriented and systems programming
- Back-end development and application logic
- Front-end development and responsive UI design
- Authentication, local caching and data handling
- Algorithms, data structures and game AI
- Machine learning and geospatial data analysis
- Professional communication and tutoring
- Fast learner with strong attention to detail
- References available on request